GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

Pythagorean Theorem in Real-World Problems

Question Bank 6.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 7, sketch the right triangle first. Round to the nearest tenth when the answer is not a whole number.

1. A TV screen is 9 inches tall and 12 inches wide. How long is its diagonal?

15 in

81 + 144 = 225.

2. A 14-ft ladder leans on a wall with its foot 3 ft from the wall. How high does it reach?

About 13.7 ft

196 − 9 = 187; √187 ≈ 13.67.

3. A ramp rises 2 m over a run of 9 m. How long is the ramp?

About 9.2 m

4 + 81 = 85; √85 ≈ 9.22.

4. A courtyard is 5 m by 13 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides?

About 4.1 m

Along two sides 18 m; across √194 ≈ 13.93 m; 18 − 13.93 = 4.07.

5. Will a 12-in ruler lie flat in a box 6 inches by 11 inches, corner to corner?

Yes — the diagonal is about 12.5 in.

36 + 121 = 157; √157 ≈ 12.53, so 12.5.

6. A kite's 13-m string is held at a spot 5 m from the point straight below the kite. How high is the kite?

12 m

169 − 25 = 144; the string is c.

7. A wire runs from the top of an 8-m pole to a stake in the ground 6 m from the pole's base. How long is the wire?

10 m

64 + 36 = 100.

Choose one.

8. A 13-ft ladder reaches 12 ft up a wall. How far from the wall is its foot?

- ○ A. 1 ft
- ● **B. 5 ft**
- ○ C. 17.7 ft
- ○ D. 25 ft

B

169 − 144 = 25 = 5^2.

9. A ramp rises 3 ft over a run of 20 ft. Which length is the hypotenuse?

- ○ A. the run
- ○ B. the rise
- ● **C. the ramp**
- ○ D. none of them

C

The square corner is under the high end; across from it is the ramp.